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RESEARCH ARTICLE

Design of a high power PIN-diode controlled switchable RF transmit array for TRASE RF imaging

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Abstract

Some MRI applications require the generation of a series high power RF pulses in which the spatial transmit B1 field pattern over the sample is modified between one pulse and the next. This requirement may be realized by a RF transmit array with the capability to enable and disable individual elements to switch between field patterns with switching times of the order 10 μ s. Our application is for a TRASE (“Transmit Array Spatial Encoding”) array for which short high power pulses are necessary to achieve high resolution (mm-level) spatial encoding. We present designs for coil array, coil switching circuits, and a high power PIN diode driver together capable of robust and rapid switching of short ($\sim 120 \mu$ s) high power pulses for a 24 cm TRASE phase gradient coil suitable for imaging extremities at 8 MHz. We describe in detail the selection of suitable coil components and switch circuit designs to satisfy a specific requirement for maximum B1 field strength, and provide all circuit designs.

KEYWORDS

PIN Diodes, RF Coil, Transmit Array Spatial Encoding

1 | INTRODUCTION

1.1 | TRASE RF encoded MRI

The Transmit Array Spatial Encoding (TRASE) MR imaging method^{1–4} achieves spatial encoding by the use of RF (B_1) phase gradients. The goal of designing practical low-cost, lightweight MR systems motivates the development of TRASE imaging and the consequent elimination of the entire conventional B_0 switched gradient system. With this goal in mind, the RF hardware described in this paper is optimized to be as compact as possible.

The TRASE imaging sequence traverses k -space by use of 180-degree refocusing pulses transmitted using different RF phase gradients. In the simplest case of 1D encoding, two different phase gradients are used alternately to

produce an echo train. TRASE echo trains traverse k -space from the center out, so that k -space data containing high-resolution information is collected toward the end of the trains. Currently, very high spatial resolution has only been demonstrated in phantom studies, using long T2 samples. In vivo, the resolution can be significantly degraded due to T2 signal decay during the echo train. The key to shorter echo trains is in improved RF technology, as discussed in this article.

Implementation of TRASE is unusual among MRI techniques in this requirement to rapidly switch between different tightly geometrically constrained RF transmit field patterns during the echo train pulse sequence. This raises some unique technical issues, as addressed here. Short RF pulses are advantageous not only for the spatial resolution reasons (described above), but also for improved off-resonance performance. Although decreased pulse length does increase power deposition, at low B_0 field strengths (here

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0.2 T), SAR constraints are much less constricting than for higher field imaging.

1.2 | Design considerations

In previous work performed using the same coil geometry as considered here,³ it was found that 300 μ s refocusing square pulses were achievable with 300 W peak RF power. In that work technical limitations of principally two types were encountered, leading to firstly thermal failures due to component overheating (PIN diodes) and secondly voltage failures (ie, sparking across capacitors). The purpose of this paper is to outline the design of a low-cost, lightweight, low power consumption PIN diode system for integration with the RF Helmholtz-Maxwell phase gradient array described in.³ We will detail the steps to determine the specifications of the transmit array such as the required input RF power, and required voltage ratings of components on the coils.

To achieve high TRASE image quality, there are a number of constraints on the operation of the collection of transmit and receive coils. The system makes use of PIN diode circuits to enable and disable the various elements. The transmit and receive coils must not interact in use, so transmit coils should be disabled when signals are to be received, and conversely receive coils should be disabled when the transmit array is active. Also during transmit any undriven elements should be effectively disabled.

The greatest factor limiting hardware development for achieving short refocusing pulse lengths is the high power requirement. First, due to “voltage magnification” in resonant LC circuits, tuning and matching capacitors require *high voltage* ratings, that is, significantly higher than the peak voltage produced by the RF amplifier. Second, each transmit element must be able to handle high *RF currents* in order to generate the B_1 field strength required for short refocusing pulses. Third, a PIN diode driver must be designed such that a sufficiently high reverse bias voltage is used to disable the coil, while supplying a sufficiently high forward bias current to ensure the diode has a low on-state resistance.

A key objective of this article is to design a PIN diode driver for the disable circuits of high power transmit array elements. The design goal is for a compact, lightweight, and low cost driver that is capable of switching 100 μ s rectangular pulses at 20% duty cycle, for a 28 cm coil. This pulse length is sufficient to allow clinical-level spatial resolution data to be acquired. This will require the driver to output a forward bias current of 500 mA or higher, and apply a large negative reverse bias voltage on the scale of -75 to -100 V. The driver will consist of three major subsystems: MOSFET switches, logic controller, and power system.

2 | THEORY

We next calculate the RF current required to drive a 28 cm square Helmholtz coil for 100 μ s, 180° refocusing pulses. From the flip angle formula, the required RF field strength is $B_1 = 117.4 \mu$ T for a circularly polarized field. For a straight wire segment (length $2a$), the magnetic field magnitude at a distance r (along the perpendicular bisector) is: $B_1(r) = 2\mu_0 I a / 4\pi r \sqrt{r^2 + a^2}$, therefore the x -directed magnetic field on axis at a distance x from the center of a square loop of 4 such wires is: $B_1(r) = \mu_0 I * 2a / \pi r \sqrt{r^2 + a^2} \sin \theta$. Using $\sin \theta = a/r$, $r = \sqrt{x^2 + a^2}$, and including a factor of 2 to account for the 2 loops of a Helmholtz coil, the current is: $I = B_1 \pi (x^2 + a^2) \sqrt{x^2 + 2a^2} / 4\mu_0 a^2$. For the 28 cm Helmholtz coil from reference³ ($a = 0.14$ m, $x = 0.12$ m) this results in: $I = 29.5$ A. Therefore, the final current required to drive 100 μ s, 180° pulses in a square 28 cm, linearly-driven, Helmholtz coil is 59.0 A.

3 | METHODS

3.1 | Coil input power

By both simulation and bench experiments, we can quantify the coil input power required to achieve the specified RF current and field. By experiment we will determine: (i) The RF voltage that appears across a capacitor on the coil, given an input power; and (ii) The RF current flowing through the coil given an input power. This gives the power required to achieve the desired RF pulse length and also the capacitor voltage ratings required. We will represent the slope of the coil current vs input voltage graph with the letter “ g ” ($g = \Delta(\text{Coil Current}) / \Delta(\text{Input Voltage})$). The current-to-voltage ratio g , gives the magnitude of the coil current that will flow when operated at a particular source voltage. The *incident voltage* will be defined as the voltage measured across the coaxial cable at the end directly soldered to the coil. The capacitor voltages vs incident voltage relationships will be represented with the letter “ G ”, and will be used to determine capacitor voltage ratings. Note that all references to voltages and currents in this experiment are in sinusoidal peak-to-peak quantities unless stated otherwise. The equipment used was as follows: Agilent E5061A Network Analyzer (determination of reflection coefficient/input impedance); Tektronix AFG 3022 Function Generator (generation of RF voltage signals at $Z_0 = 50 \Omega$); Tektronix MDO-4054B-3 Oscilloscope (RF voltage measurements); 28 cm Helmholtz coil ($Q = 220$ loaded). For the simulations, the Helmholtz coil was modeled as a circuit in Agilent ADS⁵ as shown in Figure 1. While this simulation method is not as accurate as a proper EM simulation, it provides a rough estimate to check that

the results of the experiment are in the right direction. The simulation results for input reflection coefficients and capacitor voltages are shown in Table 1 and Figure 2.

3.2 | Transmit coil input power and voltage ratings experiment

Using the function generator, sinusoidal voltages were measured across the capacitors at excitation voltages between 2 V_{PP} and 10 V_{PP} in 0.5 V_{PP} increments. As expected, a linear relationship between capacitor voltage and input voltage was observed, allowing voltage gains to be measured. From the known capacitor impedances, the current flowing through the coil was calculated by dividing the capacitor voltage by its impedance. To avoid grounding the circuit, voltage measurements were performed with two oscilloscope probes; one probe for each terminal of a component, using the subtract math function.

TABLE 1 Simulation results for RF Helmholtz coil. All peak-to-peak measurements made at 0.25 W

Parameter	Simulation result
S_{11} impedance	43.136 + j9.703 Ω
S_{11} dB	−17.929 dB
Incident Pk-to-Pk Voltage	9.336 V
Pk-pk Voltage across C1	67.74 V
Pk-pk Voltage across C2	71.01 V
Pk-pk Voltage across C3	72.07 V
Pk-pk Voltage across C4	51.13 V
Pk-to-Pk Coil Current	2.144 A

3.3 | Input power and voltage ratings experiment results

S -parameters (S_{11}) were measured on the network analyzer. S_{11} was measured at −18 dB, corresponding to

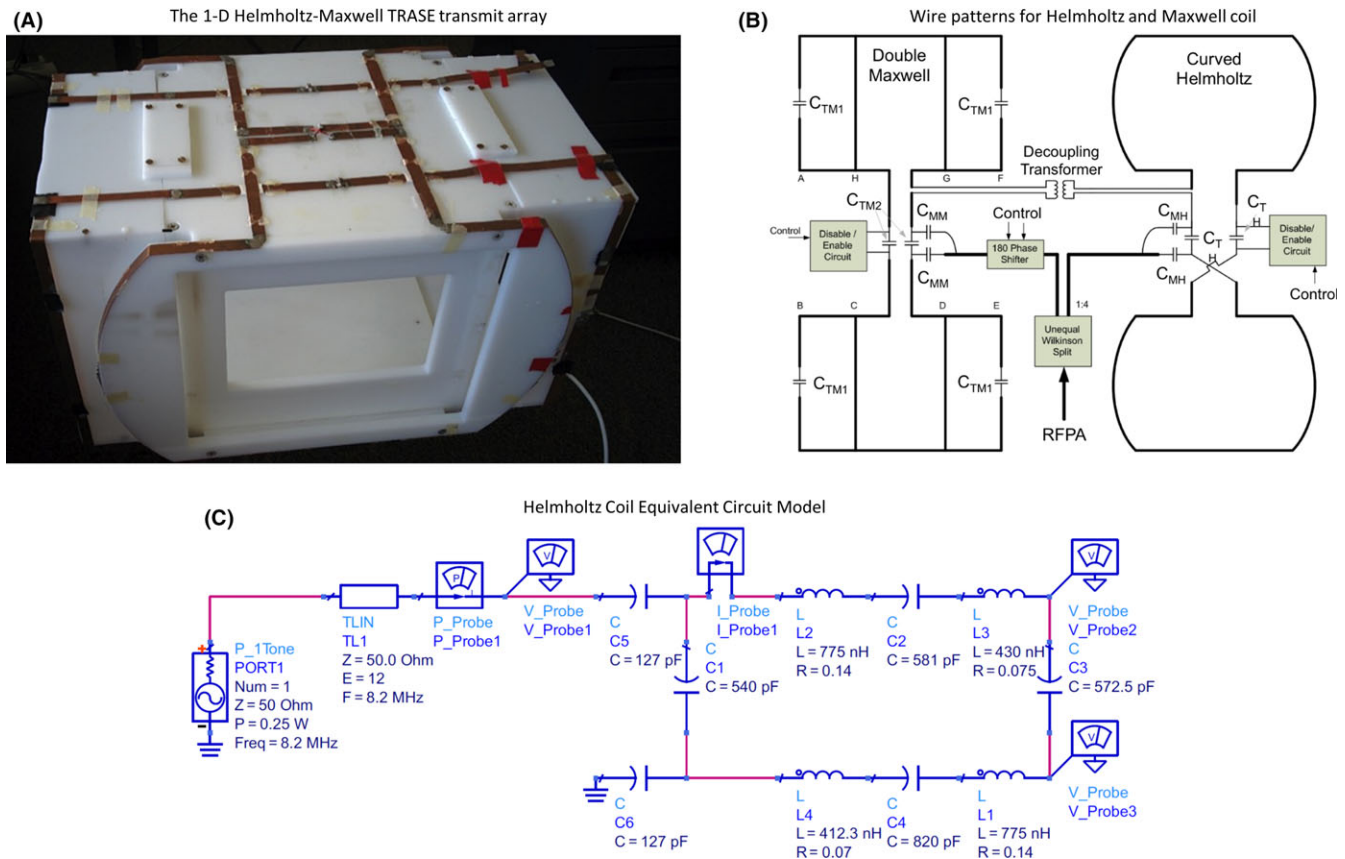


FIGURE 1 A, Photograph of the 1-D Helmholtz-Maxwell TRASE transmit array used in this study. The average spacing of the coil conductors in the imaging volume is 24 cm. B, Helmholtz and Maxwell coil wiring patterns. Capacitor values are as follows: $C_{TM1} = 1790$ pF, $C_{TM2} = 560$ pF, $C_{MM} = 45$ pF, $C_{TH} = 270 \dots 310$ pF, $C_{MH} = 57$ μ F and 72 pF. C, Helmholtz Coil Equivalent Circuit Model. Capacitor values are known, whereas the inductances and series resistances of the copper strips (modelled as inductors) were selected such that the reflection coefficient/input impedance of the coil in the simulation resembles what was measured on a network analyzer. The transmission line TL1 represents the length of coaxial cable between the RF power source and the excitation port of the coil; the electrical length of this cable is estimated to be about 12°

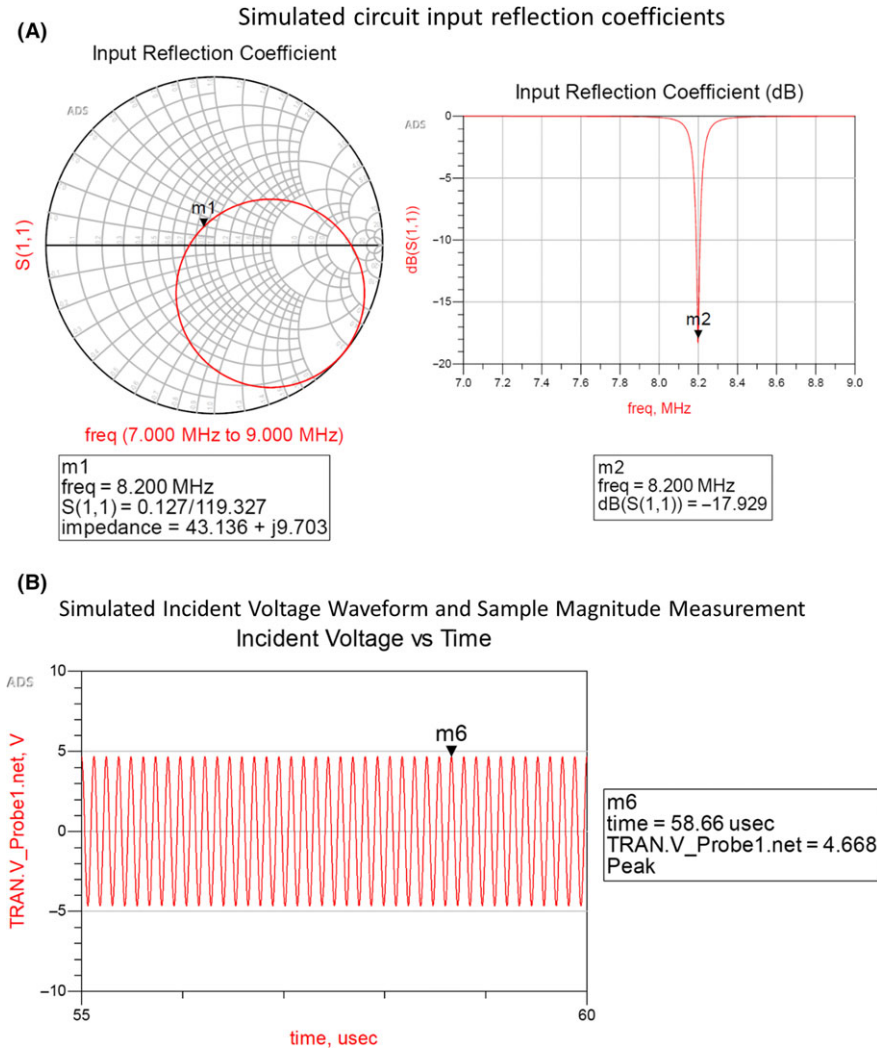


FIGURE 2 Helmholtz Coil Simulation Results. A, Simulated circuit input reflection coefficients. S_{11} impedance: $43.136 + j9.703\Omega$. S_{11} dB: -17.929 dB. B, Simulated Incident Voltage Waveform/ Sample Magnitude Measurement

an input impedance of $43.4 + j9.7 \Omega$. Voltages (V_{PP}) were measured experimentally at different power levels. The *excitation voltage* represents the voltage amplitude that the function generator is supplying at its terminals, whereas the *incident voltage* is the voltage measured across the end of the coaxial cable that is soldered onto the coil, that is, the *incident voltage* is that seen directly by the coil, once impedance mismatch and the voltage transformation due to the coaxial cable electrical length is accounted for. The simulation results indicated that the voltage is greatest in magnitude across capacitor C3. This was verified experimentally on the bench at 0.25 W. The measured voltages (V_{PP}) for C1, C3 and C4 were 63 V, 68.4 V, and 47 V, respectively. For determining capacitor voltage rating it is therefore sufficient to collect data only for C3. As expected for a linear circuit, C3 voltage was linearly proportional to incident voltage, with a slope of $G = 8.267$. This represents the maximum capacitor-to-incident voltage ratio of the coil (voltage ratio between C3 and the incident voltage).

3.4 | Analysis

First, the impedance of capacitor C3 (572.5 pF) was calculated at 8.2 MHz: $Z_C = 1/j\omega C = -j33.9 \Omega$. Then (from $I_C = V_C/Z_C$) we can obtain the peak-to-peak current flowing through the coil at varying voltage levels. (Example: $Z_C = -33.9 \Omega$; $V_C = 13.7$ V; $I_C = 0.404$ A). Coil current was confirmed to be linearly proportional to incident voltage over the tested 1 V to 8 V range. The coil current-to-incident voltage ratio for C3 can be obtained from its linear plot to be $g = 0.2439$ A/V. Finally, the incident voltage to source excitation voltage ratio was measured as 0.83. This reduction is likely due to a combination of impedance mismatch and voltage transformation due to the electrical length of the coaxial cable. This number is within reason, as it is close to the voltage absorption ratio calculated from $S_{11} = -18.0$ dB, since $(1 - 10^{-18/20}) = 0.8741$. Experimental results were within 10% of simulation predictions. The coil current-to-incident ratio g and maximum capacitor-to-incident voltage ratio ($G = 8.267$) were used to determine the power required for driving the coil and the required

capacitor voltage rating. Above we showed that a coil current of 59.0 A is required to achieve 100 μ s refocusing pulses. Dividing this by the value of g we obtained in the experiment, the incident *peak voltage* is: $V_{in} = 59.0 \text{ A} / 0.2439 \text{ S} = 241.9 \text{ V}$. From the voltage gain we can calculate the *peak voltage rating* for the capacitors: $V_C = 244.36 \text{ V} * 8.267 \approx 2000 \text{ V}$. The incident voltage to the coil is 83% of the excitation voltage applied by the source, therefore the required *peak source excitation voltage* is: $V_S = 241.9 \text{ V} / 0.83 = 291.4 \text{ V}$. In terms of RMS power: $P_S = V_S^2 / 2Z_0 = 291.4^2 / 100 = 849 \text{ W} = 59.3 \text{ dBm}$. Therefore, we can conclude that 849 W or 59.3 dBm is the RF power required to obtain 100 μ s 180° flip-angle pulses and the capacitor voltage rating should be at least 2000 V. All these results apply for a loaded coil ($Q = 220$). Coil current and capacitor voltages will both rise (relative to driving voltage) in proportion to Q .

3.5 | Diode disable circuit

In this section, we describe the design of PIN diode RF switches, used to enable and disable coil elements. In the very early history of biomedical NMR, Bendall⁶ experimented with mechanical reed relays for this purpose, which had a major disadvantage in that they could not be located within the magnetic field. In 1986, Edelstein,⁷ using some of Bendall's circuit concepts demonstrated the use of actively switched PIN diodes in MRI; see also.^{8,9} The PIN diode is a small semiconductor two terminal component suitable for RF signal control which switched between low resistance forward bias ("On") and high impedance reverse bias ("Off") states. An example of state-of-the-art use of PIN diodes to achieve very rapid switching using a feedback approach on the diode bias voltage is given by Brunner.¹⁰ PIN diodes are well described in a number of general references,^{11–14} notably including the Microsemi Corp. PIN Diode Handbook,¹⁵ which also includes high power BJT driver circuits designs (Microsemi Corp.).

Although the coil design discussed here is for a single axis of encoding (for which both transmit elements are always active), coil disable PIN diode have been implemented, as effective disable of inactive transmit coil elements is an important requirement for 2D or 3D encoding transmit arrays. The two key requirements are that inactive elements remain fully disabled, even in the presence of strong RF fields from active elements, and secondly that active transmit array elements retain high efficient (high Q) in the on-state, to efficiently deliver short RF pulses.

The most suitable PIN diode for this project was found to be the MACOM MA4PK2000. This diode has an RF and DC current rating of 25A, an exceptionally low maximum series resistance in the HF frequency range (rated 0.2 Ω at 4 MHz), and is readily available for around \$75 each.

For the TRASE sequence, the overall maximum average duty cycle is 20%, although during the echo trains the maximum is 50%. Since the RF current required for achieving 100 μ s pulses is $\approx 59 \text{ A}$ (peak); at 20% duty cycle, the average current in a series connected PIN diode is 12 A, which is within the 25 A operating range. Two schematics for disable circuits are presented (Figure 3). *Disable Circuit A* is a circuit proposed on the basis of the current handling capabilities of the diode selected for the project above and *Disable Circuit B* is a circuit previously proposed (3).

3.6 | Diode in RF path (Disable Circuit A)

From the calculation of RF current, we can show that a simple design with the PIN diode in the direct current path is prone to thermal failure. We illustrate this with *Disable Circuit A*, which introduces a PIN diode directly in series with the transmit coil's conducting path. When the diode is forward biased by a DC current, RF current can flow. When reversed bias with a DC voltage, no RF current can flow through the diode. The disadvantage of *Disable Circuit A* is that the diode may heat up to temperatures that can melt solder and the small circular geometry of the MA4PK2000 diode's package makes it extremely difficult to install a heat sink. A simple calculation illustrates this. The resistive power dissipation in the diode can be calculated as: $P = \delta I_{RF}^2 R$, where I_{RF} is the RF current, R is the effective series resistance of the diode and δ is the duty cycle of the RF pulse sequence. For 59 A current, a 20% duty cycle and worst-case 0.2 Ω effective series resistance, power dissipation is $P = 0.2 * (59A/\sqrt{2})^2 * 0.2 \Omega = 71 \text{ W}$. So, the resistive power dissipated by a series diode is higher than the power dissipation in mainstream soldering irons, and indeed in this series configuration the solder may melt and the diode fall off the circuit.

3.7 | Diode not in RF path (Disable Circuit B)

An alternative design is possible in which the diode does not see the full RF current, because the diode is in the path of a resonant trap circuit with relatively high reactance components (see Figure 3b, and ref. [3]). The RF current circulating inside the resonant trap is much lower than the 59A RF current that direct design must handle. In detail, when the diode is reverse biased, the 4.79 μ H inductor and 91 pF forms a resonant trap with the 560 pF tuning capacitor on the transmit coil at the frequency of interest, not allowing any RF current to flow through the coil at the frequency of interest (transmit coil is detuned). When the diode is forward biased with a DC current the 3.9 μ H inductor in the diode branch forms a resonant RF trap with the 91pF capacitor in the branch below it, that branch can

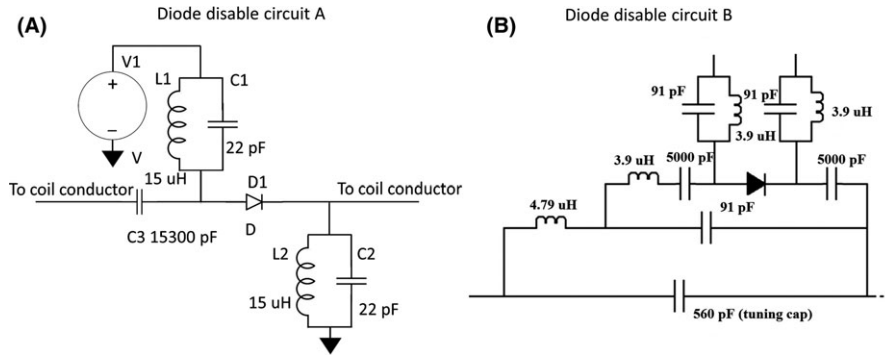


FIGURE 3 TRASE transmit coil and diode disable scheme topologies. A, Disable Circuit A, with diode directly in the RF current path; B, Disable Circuit B, with diode protected from the full RF current

no longer resonate with the tuning capacitor, the transmit coil is resonant and RF current can therefore flow through the transmit coil. The LC circuits in the wires used to drive the PIN diode resonate at 8.2 MHz and so present a high impedance to prevent RF currents flowing on the DC supply wires. The magnetic field strength of the 0.2 T permanent magnet can vary seasonally depending on the temperature of the magnet suite, which can result in impaired trap performance due to the drift in Larmor frequency.

3.8 | Other components

Based on the 2000 V calculation, capacitors suitable for the transmit array are American Technical Ceramics 100C and 100E series which have voltage ratings up to 2500 V and 3600 V, respectively. The receive coil requires a separate logic control channel from a PIN diode driver. The receive coil is disabled when the diode is forward biased and enabled when reverse biased. This is contrary to the T_X coils, which are disabled when reverse biased and enabled when forward biased. The system is designed to operate from a single RF power amplifier, so a power splitter is required to drive both transmit coil elements simultaneously. The direction of the phase gradient is reversed by a 180-degree phase shift in the driving signal to the Maxwell coil. This was implemented by installing a custom-built 8.2 MHz, 50 ohm, 180-degree phase shifter after the power splitter. The phase shifter design consists of two alternate parallel RF paths, only one of which is active at one time. Each path contains a PIN diode switch (each consisting of two back to back diodes), so two separate PIN diode drive lines, driven with complementary logic signals, are required. The upper path contains two high-pass T circuit elements in series, each providing a 90-degree shift, for a total of 180°. The lower path contains a high-pass T element followed by a low-pass PI element for a total shift of 0°.

4 | DIODE DRIVER DESIGN

PIN diodes switches are required for both coil enable/disable function and also for the control the control of the RF phase shifter. Our design was based on Xiao's¹⁶ MOSFET drivers, rather than BJT designs,¹⁵ for several reasons. MOSFETs generally have faster switching speeds, lower switching losses and are suitable for higher voltage designs. For our transmit array application different transmit elements are not necessarily well geometrically decoupled, so a high voltage is needed to maintain a robust off state. The MOSFET design is also simpler than a BJT design, in addition to the logic and power circuitry needed to drive bipolar transistors, there are many extra components in the BJT design not required in Xiao's design. Transistors are switched using optocouplers and low power voltage regulators in our design, and we are able to do so because MOSFETs are voltage driven devices where the gate terminal does not draw current. Were a BJT design used instead, a larger more powerful converter would be needed. Our diode driver design is therefore based on Xiao's MOSFET push-pull scheme, but adapted to drive a low frequency (8 MHz) transmit array, rather than a transmit-receiver switch.

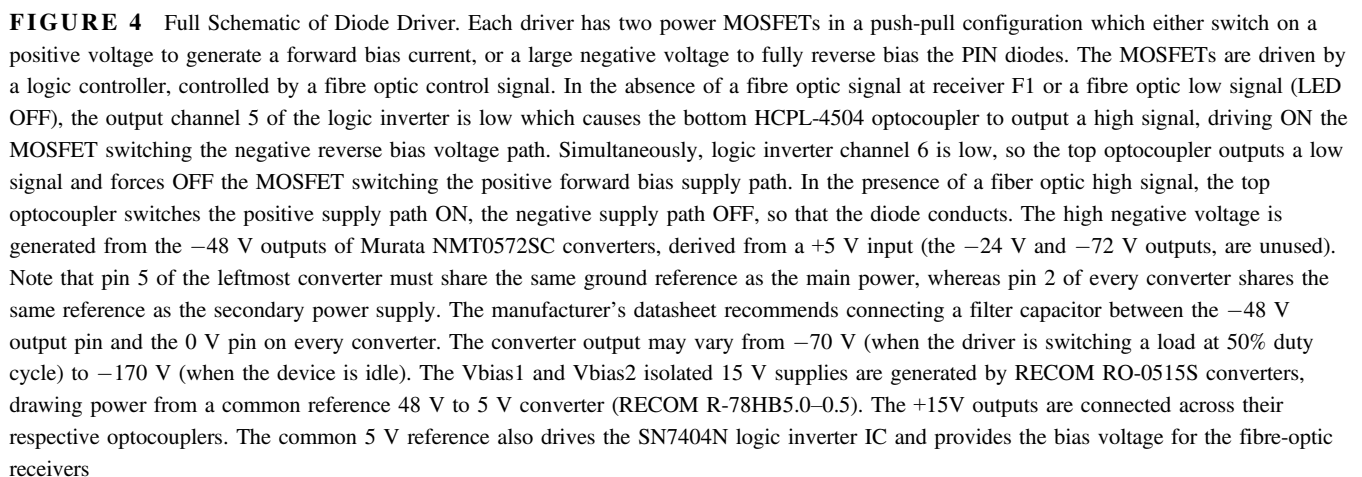
We describe in this section the design of the PIN diode driver circuitry, consisting of three major parts: MOSFET switches, logic controller, and power system (Figure 4). The diode drivers are controlled by fiber optics signals to prevent the transmission of electrical noise into the magnet room. Four boards of the diode drivers were constructed for the four different logic channels required for the 1D imaging system. The design of the PIN diode driver as a stack of boards allows a simple extension to more channels as needed for higher dimensional imaging applications, see Figure 5.

4.1 | MOSFET push-pull switch

The design consists of two power MOSFETs that are switched with opposing states¹⁶; for example, when

constants. Therefore, each channel has a series resistance of 82 Ω to ensure the RL time constant defined as $\tau = L/R$ seconds is small for a range of inductive RF chokes and allows for fast forward biasing switching. In addition, the resistors limit the steady-state current to 580 mA.

From Figure 4, it can be seen that two control signals of opposite logic are required to switch between the positive and negative supplies of the device. However, the gate-



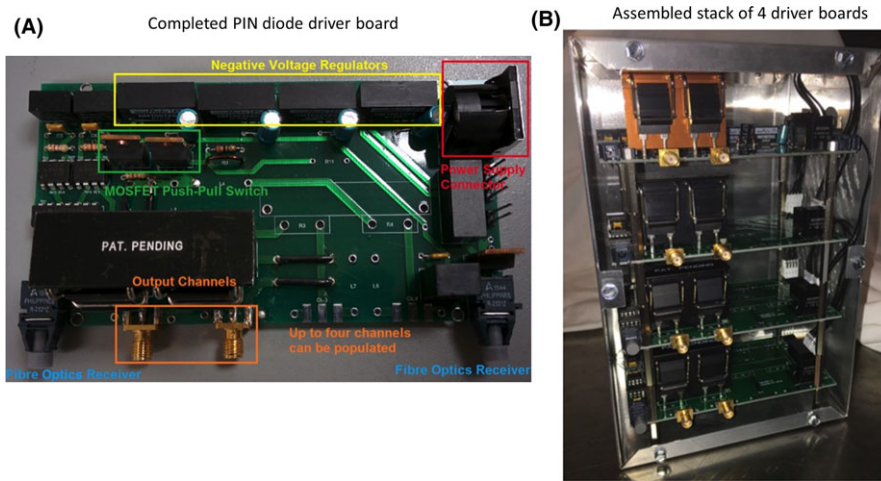


FIGURE 5 Photographs of completed hardware. A, PIN diode driver board. The diode drivers are controlled by fiber optics signals to prevent the transmission of electrical noise into the magnet room. Each driver board has two power MOSFETs which act as switches to either switch on a positive voltage to generate a forward bias current or a large negative voltage to thoroughly reverse bias the PIN diodes. Up to four output channels can be populated on the diode driver, controlled by a single fiber optic control signal. The diode driver board dimensions are 13.5×7.2 cm; B, Assembled stack of 4 driver boards

source voltage of the MOSFET must be around 15 V in order switch the MOSFET on, considerably higher than TTL outputs voltages, between 3.3 and 5 V. Therefore, a sub-circuit was designed to step low voltage control signals to 15 V pulses for driving the gate. In addition, one control signal must have the inverted logic of the other; therefore a logic inverter is required. The schematic in Figure 4 depicts the logic controller sub-circuit. The Avago HCPL-4504 Optocoupler is used to drive the MOSFET gates from a single TTL input (FO signal converted to TTL by HFBR-2521Z Receiver). Two channels of a logic inverter (TI SN7404N) are used to set the opposite states of the MOSFETs in the push-pull switch. V_{BIAS1} and V_{BIAS2} must be isolated 15 V supplies in order to independently control the gate-source voltages of the MOSFETs.

4.3 | Power system

The main power supply, V1 is a 48 V 5A DC supply, the size of a laptop charger and can be connected to the wall AC socket. The negative voltage supply is implemented using four isolated -48 V DC-to-DC converters. Two pairs of parallel connected -48 V converters are connected in series. Each converter takes a +5 V input and steps the voltage down to -48 V, see Figure 4. The voltages V_{BIAS1} and V_{BIAS2} for driving the MOSFETs' gates are also implemented using isolated DC-to-DC converters. Two additional issues need to be addressed: First, the negative voltage converter makes use of multiple isolated DC-to-DC converters that must share a separate input reference from the main supply; second, the high voltage supply is ON even in the absence of a fiber optic control signal, this can

be a potential safety hazard. As a result, additional fiber optic control was implemented as a safety precaution to disconnect all high voltages, when in the idle state; a controllable 48 V to 5 V isolated converter (RECOM RS3-4805S/H2) was selected to power the four negative voltage converters.

4.4 | Transmit coil power tests

Figure 6 shows the experiment setup for transmitter coil bench power tests. The maximum duty cycle that can be used in the TOMCO BT00500 power amplifier is 20%; therefore all power tests were ran at 20% duty cycle at the RF pulse RMS power stated (eg, 400 W, 500 W etc.) for 30 minutes, or until failure (if sooner).

5 | RESULTS

5.1 | PIN diode driver test results/switching speed and power consumption results

To evaluate the switching speed of the PIN diode driving system, a test circuit was constructed. It contained a 4.7 μ H inductor representing the inductor used in the RF traps on the diode lines, in series with a 1 Ω resistor for current measurement, and the PIN diode under test. The switching speed was evaluated by measuring the bias voltage waveform across the diode, evaluated as the elapsed time between 0% and 90% of the transient. Current was measured by monitoring the 1 Ω resistor voltage. Two different diodes were used to evaluate the driver, the MACOM MA4PK2000 and the Microsemi UM9415SM. The

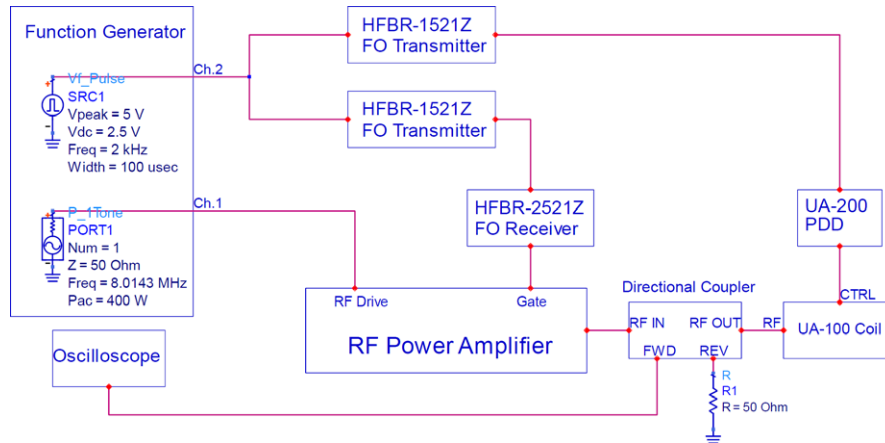


FIGURE 6 Transmit coil bench power test setup. The transmitter circuit is simple and involves placing a single $47\ \Omega$ ballast resistor in series with the HFBR-1521Z fibre optic transmitter IC. The fibre optic transmitter sends the control signals generated by the function generator to the TOMCO BT00500 500 W power amplifier and the PIN diode driver. The PIN diode driver forward biases or reverse biases the PIN diode on the disable circuit

MA4PK2000 has a slower reverse bias switching speed as a result of a longer carrier lifetime. Pulses were generated using a function generator (Tektronix AFG 3022), while all waveforms measurements were performed using an oscilloscope (Tektronix MDO4054B-3). The measurements of the overall PIN diode waveform and detail of diode voltage rise and fall time are plotted in Figure 7. Table 2 summarizes the switching speed and current measurements. Both rise times were satisfactory and within $1\ \mu\text{s}$, while fall times vary from $3.8\ \mu\text{s}$ to $7.4\ \mu\text{s}$.

Some additional tests were also performed, neither of which resulted in significantly affected rise or fall times. First, a modified driver board with a lower reverse bias of $-82.3\ \text{V}$ was used. Second, the inductor was changed from $4.7\ \mu\text{H}$ to $68\ \mu\text{H}$. The $68\ \mu\text{H}$ test case is to ensure the driver design is robust, that is, that it is capable of switching a high inductive load in less than $5\ \mu\text{s}$. Although fall and rise times were not affected, we observed a video leakage bias ringing of $5.6\ \text{MHz}$ in the fall time region for the UM9415SM diode with the $68\ \mu\text{H}$ inductor. This is consistent with the UM9415SM being the faster diode, and so more prone to this effect. This ringing frequency of $5.6\ \text{MHz}$ is close to our NMR operating frequency ($8.2\ \text{MHz}$) and therefore difficult to filter out. For all coil experiments the MA4PK2000 was used which did not exhibit ringing, as evaluated by the bias voltage across the diode.

Diode driver board power consumption was evaluated. The idle-state power consumption for one diode driver board is $3.36\ \text{W}$. For a single diode load of $580\ \text{mA}$, and $48\ \text{V}$ the power consumption is $17.3\ \text{W}$. Each additional diode branch adds $13.92\ \text{W}$ to this value. The $82\ \Omega$ resistor heats up significantly due to it dissipating about $13.92\ \text{W}$ in the current on state.

5.2 | Coupling between coil elements

Coupling between coil elements is unwanted and potentially could lead to the energization of a disabled coil. We found that in the worst-case scenario, there is an S_{21} of $-20\ \text{dB}$, between the BNC ports of the two transmit elements when one element is disabled and the other one is transmitting. The maximum power to be transmitted in a TRASE experiment is about $1000\ \text{W}$. Therefore, $10\ \text{W}$ of power is observed in the disabled coil's BNC port. Simulation on Keysight ADS by applying $10\ \text{W}$ to a disabled coil shows that a maximum of about $6.6\ \text{V}$ appears across the tuning capacitors. Calculation of the voltage division across the components on the resonant disable circuit, shows that about $6.6\ \text{V} * 7.6 = 50.16\ \text{V}$ appears across the diode. Therefore, a reverse bias voltage of $-96\ \text{V}$ is more than sufficient for ensuring the disabled coil remains off.

5.3 | Transmit coil power test results

RF power bench tests were performed using a TOMCO BT00500 RF power amplifier (RFPa) at $400\ \text{W}$ and $500\ \text{W}$. An initial 2-minute $400\ \text{W}$ coil stability test was performed, after which the coil was immediately connected to a network analyzer. It was noted that the coil's resonant frequency (frequency at the valley of the S_{11} curve) shifted down by $15.7\ \text{kHz}$. The coil matching also dropped from $-20\ \text{dB}$ to $-8\ \text{dB}$. For further tests, the coil was retuned so that the coil was tuned and matched at the working temperature, and a 30-minute power test was conducted. For the $500\ \text{W}$ power tests the same pattern was observed, but with a $20\ \text{kHz}$ shift. No arcing was observed on any capacitors, diodes remain intact on its circuit, but capacitors and diodes heat up significantly after the tests.

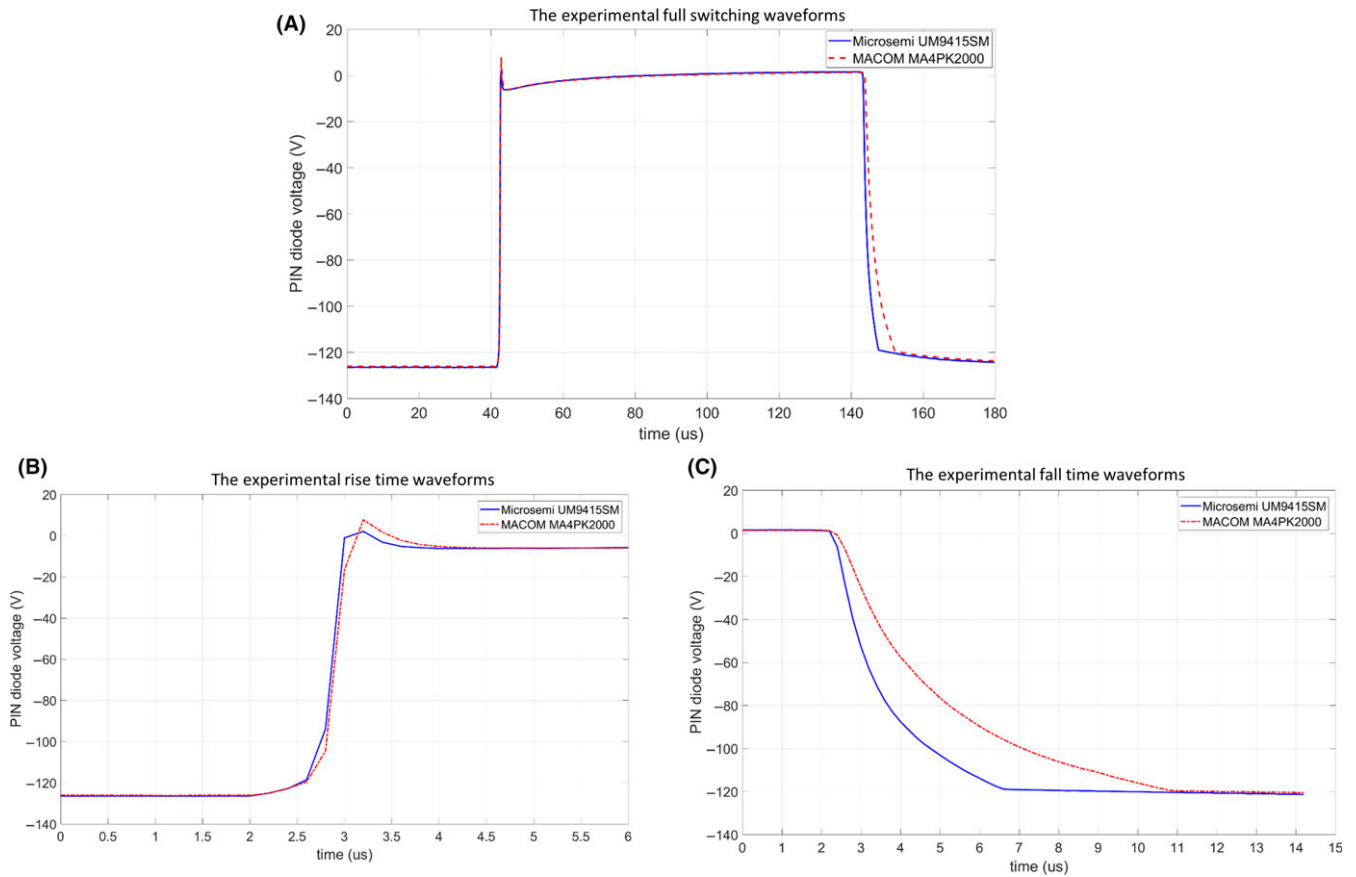


FIGURE 7 Diode Driver Switching, Experimental Results. The switching speed of the driver system was evaluated based on the high voltage board configuration. The rise and fall times of the voltage applied to the driven PIN diode were recorded using the difference voltage between two oscilloscope probes attached either side of the diode under test. Two different PIN diodes: the MACOM MA4PK2000 and the Microsemi UM9415SM were tested under the same experimental set-up. Data were analyzed in MATLAB. A, The full switching waveforms showing both rise and fall times for PIN diode voltage. B, Detail view of rise times. The rise time of two different measurements were both 0.9 μ s, and were independent of the diode selection. C, Detail view of fall times. The MA4PK2000 exhibited a fall time of 7.4 μ s, which was longer than the fall time (3.8 μ s) of Microsemi UM9415SM. The two diodes have different carrier lifetimes, and the MA4PK2000 has a slower reverse bias switching speed as a result of a longer carrier lifetime, hence longer fall time

A flip-angle experiment was performed on the 0.2 T magnet using the array's Helmholtz coil, the results are as follows. RF pulses of 361 W were required to achieve a 180° condition for a pulse length of 180 μ s, while 940 W pulses were required to achieve the 180° condition in 100 μ s. The tests used a very low duty cycle, therefore the coil components did not heat during operation. No arcing of the coil tuning capacitors was observed in the full duration of the experiment.

5.4 | Noise measurements

The connection of the diode control lines has the potential to introduce additional noise. This was assessed by acquiring data with the spectrometer with the RF amplifier switched off, with the diode control line first connected and secondly disconnected. This is possible because the receive coil remains tuned when the diode is not driven.

With the diode line connected higher noise was observed (25 dB). A pi Butterworth low-pass filter ($L1 = 27 \mu$ H, $C1 = C2 = 2000$ pF, 3 dB = 1 MHz) was used in series with the diode control lines. In conjunction with using a common ground for the RF and diode grounds this reduced added noise to negligible levels. The noise attenuation at 8.2 MHz due to filtering is approximately -60 dB. This is sufficient for use at 0.2 Tesla, however, for very low field systems (eg, 0.05 Tesla) additional attention to noise reduction will likely be necessary, especially for receive coils. The computer brick power supply was compared to a (larger and heavier) linear bench power supply. Noise levels were the same in both cases. A 25% reduction in noise voltage was observed by situating either power supply outside the magnet room, with the 48 V voltage power line passing through the filters in the filter panel. We did not investigate whether in-room filtering could achieve the same reduction. As a general comment, correct grounding

TABLE 2 Experimental results for switching speeds for two PIN diode models. The switching speeds were evaluated at 50% duty cycle using 100 μ s pulses and with the 82 ohm load resistor in circuit

PIN diode type	Microsemi UM9415SM	MACOM MA4PK2000
Reverse bias (V)	−125.4	−123.6
Rise starting bias (V)	−125.4	−123.6
Rise ending bias (V)	−11.5	−11.3
Rise time (μ s)	0.9	0.9
Fall starting bias (V)	1.4	1.3
Fall ending bias (V)	−110.9	−110.6
Fall time (μ s)	3.8	7.4
Forward bias (A)	1.63	1.72

is critical to avoid introduction of extraneous noise, and inevitably the introduction of diode driver cables to the coils make grounding issues more complex.

5.5 | One-dimensional TRASE imaging results

The spatial encoding capability of the coil array was verified in a 1-dimensional TRASE phantom imaging experiment using a cylindrical vial of water (32 mm internal diameter height 20 mm), see Figure 8. The pulse sequence consisted of a single echo train of 128 refocusing pulses of length 200 μ s. Although the coils can function with 130 μ s pulses, in this experiment the performance of the phase shifter circuit limited the pulse length to 200 μ s. The figure shows profiles both with and without phase switching, illustrating the point spread function and a 1D profile, respectively.

6 | DISCUSSION

The main conclusion of this work is that, using the described PIN diode driver, switch circuitry and RF coil array, short RF pulses can be produced at high power without significant distortion. The circuit simulations were accurate to within 10%, which is more than sufficient for design purposes, including the selection of the appropriate components. The improved capacitor and diode circuitry resolved hardware limitations of the previous implementation, including capacitor arcing and thermal failure.

This new capability to operate at higher power levels for extended pulse sequence durations does uncover new thermal stability issues, which we discuss briefly here. The power tests revealed that coil resonant frequency was susceptible to drift with temperature, which is related to temperature dependence of the capacitors. The datasheets of

the ATC 100C and 100E series indicate up to a 0.5% capacitance increase for a 40°C temperature increase. An increase in the capacitance of the tuning capacitors of the RF transmit array, results in a decrease in the resonant frequency ($\omega = \frac{1}{\sqrt{LC}}$) which agrees with the increased resonant frequency observed in the bench power test. In future work, a further upgrade to use capacitors with dielectrics with more stable thermal characteristics will be pursued. For example, the ATC 700C, 700E, 800C, or 800E series of capacitors are rated as five times more stable than the 100C and 100E series.

Finally, we discuss two limitations to the scope of this work. First, we did not directly experimentally compare the noise performance of this compact switched power supply design with that of a linear design. Such comparisons might be useful in more clearly distinguishing between noise generated within the driver itself and noise due to grounding issues.

Second, we comment on switching speed. The system as described is adequate to allow 100–120 μ s RF pulses with a dead time of 40 μ s before acquisition with no

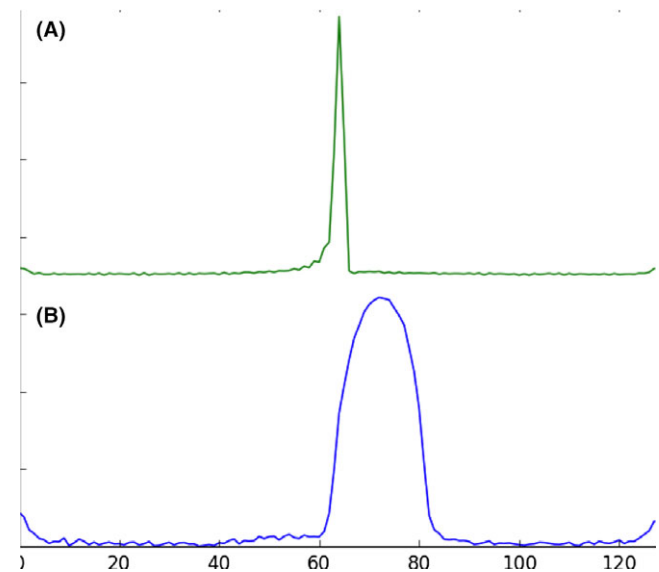


FIGURE 8 TRASE 1D Imaging Results. This figure shows two datasets obtained by FFT of TRASE-reordered echo data from a single echo train, using a small cylindrical vial of water (32 mm internal diameter height 20 mm) as phantom. A, This plot was acquired with spatial encoding disabled (ie, no coil switching), so the sharp peak approximates the point spread function (PSF). The finite width indicates T2 relaxation and the finite length of the echo train. B, This plot was acquired with 1D TRASE encoding enabled and shows a 1D profile of the phantom. Both datasets were collected with the following parameters: 1 average, 128 echoes, TE = 1200 μ s, pulse length 200 μ s, readout 16 points at 20 kHz bandwidth. Delays of 40 μ s both before and after the refocusing pulses were used. For display purposes the PSF has been scaled down by a factor of 5.5

discernable switching leakage in the acquisition window. Moderately improved performance is likely achievable in some cases; in our echo train application there is more tolerance to initial glitches than for FID type measurements as the NMR signal is formed further from the RF pulse and symmetry of the echo signal may be used to recover corrupted data. However, significant improvement in switching speed would have to address other issues including increased video leakage from the switch,¹⁰ and coil ringing, which is more prominent for low field MRI due to the longer decay time.

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